

Using Large Language Models for Evaluation of Radiological Textual Reports

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Outline

- ***Nature of Medical Imaging***

- Necessity of various imaging schemes in radiology
- Contribution of Natural Language Processing

Methodology of Automatic Report Assessment with GenAI

- **Experiments Organization**

How well textual LLM are capable to retrieve image acquisition aspects from textual narratives metadata

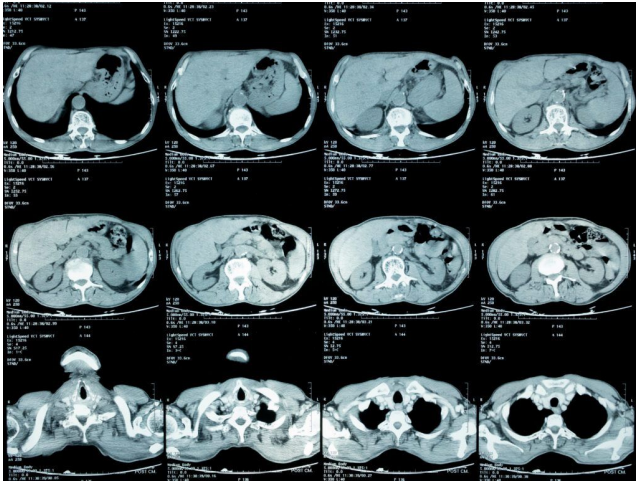
DEMO

Introduction

Introduction into Medical Imaging

Examining Patient Conditions

- Methods MRI / CT / Ultrasound
- **Output: Signal**
- Signal varies in quality depending on method.



Surveillance Challenges

- **Signal -> Images**
- Requires Analysis of various imaging schemes from collected signal

Sources of information in Medical Imaging

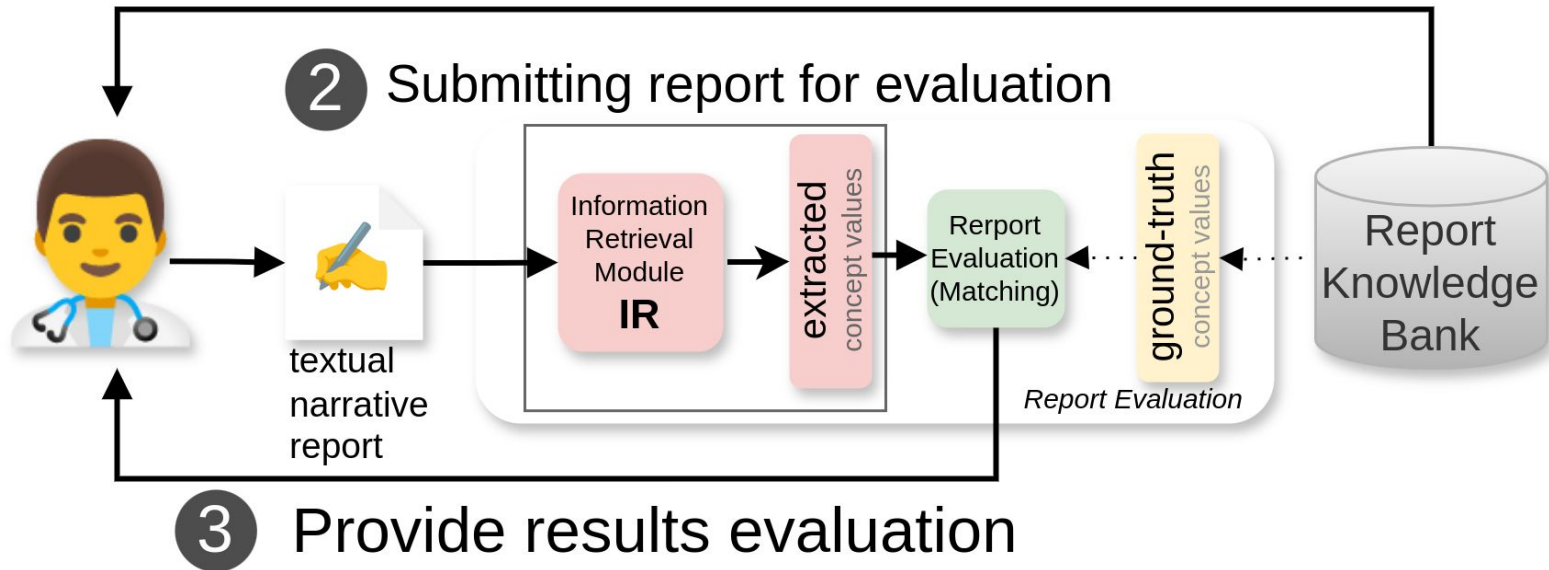
- ⚙️ Metadata
- Preparation of the patient
 - ✍️ **Textual narratives**



The role of **Information Retrieval** in Report Evaluation

① Extracting data for organizing training case

② Submitting report for evaluation



③ Provide results evaluation

Methodology

Methodology

T — task provided in description form and set of the expected closed-end answers (**C**).

I — patient info

Ap — practitioner textual narrative,

Ag — ground truth answer in **C**,

S — list of explanation categories.



SCHEMA:



$$\mathbf{I}_M(T, A_p, C) \rightarrow A'_p$$

$$\mathbf{I}_M(T, A_p, A'_p) \rightarrow F_p$$

$$\mathbf{I}_M(T, I, A_g) \rightarrow F_g$$

$$\mathbf{I}_M(T, A'_p, A_g, F_p, F_g, S) \rightarrow E$$

Mark patient conditions according to LiRADS-2018

Imaging has sing of tumor and hepatocellular carcinoma

SUBMIT

Recognized Answer

The response is classified as LR-5.

Result

correct

Explanation

The alignment of the response "The response is classified as LR-5" to the ground truth answer LR-5 can be mapped by matching the practitioner facts (2) onto the ground truth facts (1) with respect to LI-RADS key aspects.

Imaging Features

The practitioner fact "Imaging has sign of tumor" corresponds to the ground truth fact "Bounding box (T2WI)" which suggests that the T2-weighted image (T2WI) shows a lesion with high signal intensity, a characteristic feature of LR-5.

Experiments

Automatically concepts retrieval
from medical textual narratives

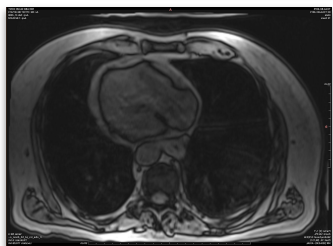
Analyzing existing reports on Medical Images



DICOM metadata

Limited DICOM attributes All DICOM attributes			
Search: Descr			
Tag ID	VR	Tag Name	Value
(0008,1030)	LO	StudyDescription	MRI ABD W+WO CONT
(0008,103E)	LO	SeriesDescription	t1_vibe_tra_p2_bh_dixon_opp_FIL
-->(0008,0104)	LO	CodeMeaning	Clean Descriptors Option
(0032,1060)	LO	RequestDescription	MRI ABD W+WO CONT
(0040,0254)	LO	Performance	

	G	H	I
7	MRI ABD WWQ CONT	06-06-2000	t2hasteTshicksIp2
8	MRI ABD WWQ CONT	06-06-2000	t1ftraTE2.3
9	MRI ABD WWQ CONT	06-06-2000	t1vibeqfstrap2bhFIL
10	MRI ABD WWQ CONT	06-06-2000	t1vibeqfstrap2bhFIL
11	MRI ABD WWQ CONT	06-06-2000	t1vibeqfscORp2bhFIL
12	MRI ABD WWQ CONT	06-06-2000	t1ftraTE4.1
13	MRI ABD WWQ CONT	06-06-2000	T1 FL AX IN-OUT PHASE
14	MRI ABD WWQ CONT	06-06-2000	t1vibeqfstrap2bhFIL
15	MRI ABD WWQ CONT	02-21-2000	T1FLASHCOR
16	MRI ABD WWQ CONT	02-21-2000	t2tseCORCBD
17	MRI ABD WWQ CONT	02-21-2000	NEW HASTE CORONAL
18	MRI ABD WWQ CONT	02-21-2000	t1vibetrap2bhcdixonoppFIL
19	MRI ABD WWQ CONT	02-21-2000	NEW HASTE AXIAL SPAIR
20	MRI ABD WWQ CONT	02-21-2000	t1vibetrap2bhdixonFFIL
21	MRI ABD WWQ CONT	02-21-2000	t1vibeqfscORp2bhFIL
22	MRI ABD WWQ CONT	02-21-2000	t1vibeqfstrap2bhFIL
23	MRI ABD WWQ CONT	02-21-2000	t1vibetrap2bhdixoninFIL
24	MRI ABD WWQ CONT	02-21-2000	t1vibeqfstrap2bhFIL
25	MRI ABD WWQ CONT	02-21-2000	IN-OUTPhaseAxial
26	MRI ABD WWQ CONT	02-21-2000	t1vibeqfstrap2bhFIL
27	MRI ABD WWQ CONT	02-21-2000	t1vibetrap2bhdixonWFIL
28	MRI ABD WWQ CONT	05-08-1998	NEW HASTE CORONAL



XLSX-metadata

Technique:

opposed
phase /

out-of-
phase

Technique:
spin-echo



t1ibetrap2bhdixonoppFIL

Imaging:
T1-
weighted
series

Imaging:
Fat /
Water
Separation

<https://www.cancerimagingarchive.net/collection/tcga-lihc/>

Dataset ($SN_{HCC-TCIA}$)

We adopt **pattern-matching rules** for extracting parameters related to each **atomic aspect** of three categories

IMAGING

↓
Weighting

↓
T1 T2
DWI ADC

CONTRAST

↓
Timing

↓
YES
NO

ACQUISITION

↓
Plane

↓
axial
sagittal
coronal

<https://github.com/nicolay-r/Reasoning-for-Radiology-Report-Evaluation>

Results

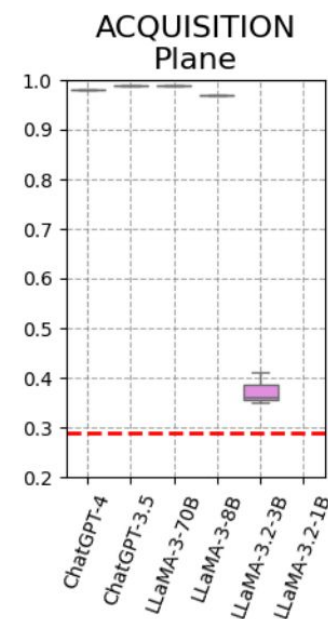
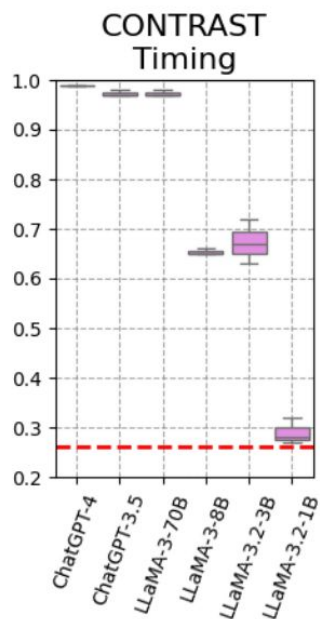
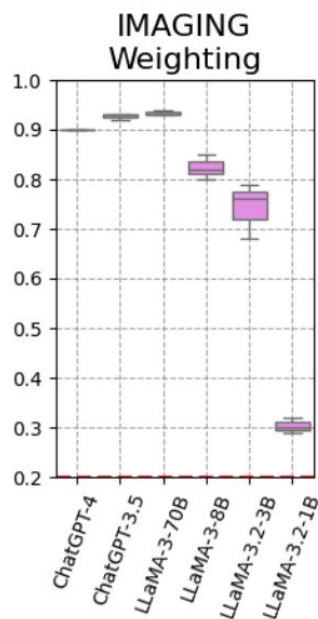
Evaluation Metric: F1 over all the classes of the particular category (except unknown)

- For the Fat Suppression concept we adopt **Accuracy**

Method	WEIGHTING	TIMING	PLANE
ChatGPT-4 (1.76T)	0.90	0.99	0.98
ChatGPT-3.5 (175B)	0.93	0.97	0.99
LLaMA-3-70B	0.93	0.98	0.99
LLaMA-3-8B	0.85	0.65	0.97
LLaMA-3.2-3B	0.68	0.72	0.36
LLaMA-3.2-1B	0.32	0.27	0.18
RANDOM	0.19	0.26	0.29

<https://github.com/nicolay-r/Reasoning-for-Radiology-Report-Evaluation>

Key findings out of organized experiments



Findings:

- **Open-transparent LLaMA-70B performs similar** to ChatGPT systems
- **Noticeable performance drop** in the case of small LLMs (3B and less)
- small LLMs illustrate less confident behaviour

Example. User interface for

The screenshot displays a medical imaging software interface. The main area shows a cross-sectional MRI scan of a liver. On the right side, there is a panel titled 'Surveillance Task' which includes a 'Mark patient conditions according to LI-RADS-2018' section. Below this, there is a 'SUBMIT' button and a 'Recognized Answer' section stating 'The response is classified as LR-5.' Further down, a 'Result' section shows 'correct' in a green box, followed by an 'Explanation' section and an 'Imaging Features' section. Three callout boxes are overlaid on the interface: a light blue box on the left points to the main image area, a yellow box on the right points to the 'SUBMIT' button, and a green box at the bottom right points to the 'Results' section.

3. Area for viewing studies by practitioner

2. Area for results submission

3. Results

Thank you for your attention!



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